MIL-T-14605A(MU)
30 November 1967
SUPERSEDING
MIL-T-14605(CmlC)
23 September 1957

MILITARY SPECIFICATION

TRANSFORMER UNIT

1. SCOPE

1.1 This specification covers one type of electrical transformer.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Fiberboard.
PPP-T-60	-	Tape; Pressure Sensitive Adhesive, Waterproof - For Packaging.
QQ-S-781	-	Steel Strapping, Flat.

MILITARY

MIL-P-116	-	Preservative, Methods of.
MIL-B-131	-	Barrier-Material; Water Vaporproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

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DRAWINGS

US ARMY MUNITIONS COMMAND

DL5-66-274 - Transformer Unit.
D150-2-1 - Marking Diagram for Shipping Containers.

(Copies of specifications, standards, and drawings, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL5-66-274 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the transformer unit shall conform to the specifications and drawings listed on Drawing DL5-66-274 and subsidiary drawings.

3.2 Manufacture and assembly. All components of the transformer unit shall be manufactured and assembled in accordance with their respective assembly drawings listed on Drawing DL5-66-274.

3.3 Performance. The transformer unit shall have a voltage output of 25 ± 1 volts at 5 amperes with 115 volts 60 cycles per second input, and shall show satisfactory functioning of the relay when tested as specified in 4.2.4.

3.4 Workmanship. All metal parts of the transformer unit shall be free of burrs and sharp edges. Wiring shall be free of kinks and cuts, breaks, and abrasions of the insulating surface. All parts shall be free of contamination such as dirt and foreign mat-

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for tne performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Regular production.

4.2.1 Lotting. A lot shall consist of the transformer units produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

4.2.2 Sampling for examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection.

4.2.3.1 For examination. The transformer units shall be examined in accordance with the classification of defects and MIL-STD-105.

4.2.3.2 For critical defects. Each transformer unit in the lot shall be tested for critical characteristics listed in the classification of defects.

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4.2.3.3 Classification of defects.

(a) Transformer Unit (Drawing E5-66-274).

Categories	Defects	Acceptance standards
Critical:		
1	Performance	4.2.4
Major:	AQL 1.0 percent defective	
101	Components missing	
102	Components incorrectly assembled	
103	Basic dimensions	
104	Cuts or breaks in insulation	
Minor:	AQL 4.0 percent defective	
201	Protective finishes	
202	Workmanship (3.4)	

4.2.4 Performance. Each transformer unit shall be tested at room temperature as follows: Connect the "lead to purifier" to a variable 10 ohm (200 watt minimum) variable load resistor with an alternating current voltmeter and ammeter connected to measure line current and voltage.

Connect the "lead to 115V 60 cycle supply" to a source of 115 volt, 60 cycle, single phase, alternating current.

Flip the toggle switch to the "on" position.

Observe the ammeter reading and adjust, if necessary, the variable load resistor to obtain a 5 ampere output. Observe voltmeter for compliance with the 25 ± 1 volt requirement of 3.3. (If voltage is acceptable proceed to the next operation).

Connect the "lead to 24V Battery" to a source of 25 ± 1 volt at 5 amperes 60 cycle, single phase, alternating current.

Disconnect the "lead to 115V, 60 cycle supply."

Observe voltmeter and ammeter; readings of 25 ± 1 volt and approximately 5 amperes respectively, indicate satisfactory operation of the relay; flip toggle switch to "off" position.

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Observe voltmeter and ammeter; readings of zero indicate satisfactory operation of toggle switch.

Disconnect the "lead to 24V Battery." Repeat the cycle four more times.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A. One transformer shall be packaged in a fiberboard box conforming to style RSC, type SF, class weather-resistant of PPP-B-636. The electrical cable shall be coiled and secured to the top of the unit (see Drawing C5-66-278). Cells shall be fabricated and positioned about the unit as shown on Drawing C5-66-278. Filler pads, fabricated from the same material as the cells, shall be used as required to insure a tight pack. The box shall be closed and sealed in accordance with the appendix thereto and the entire package shall then be packaged in accordance with method 1A8 of MIL-P-116.

5.2 Packing. Level shall be as specified (see 6.2).

5.2.1 Level A. The transformer unit, packaged as specified in 5.1, shall be packed in a style 4, nailed wood box, conforming to PPP-B-621. Fiberboard pads, type CF, class domestic, of PPP-B-636, shall be placed at the top, bottom, sides and ends of the box. The box shall be closed in accordance with appendix of PPP-B-621. Strapping conforming to type I, class B of QQ-S-781 shall be used.

5.2.2 Level C. The transformer, packaged as specified in 5.1 shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification Rules or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or orders, unit packages shall be marked in accordance with MIL-STD-129. Shipping containers shall be marked in accordance with D150-2-1 (Marking Diagram for Shipping Containers).

6. NOTES

6.1 Intended use. This specification covers a transformer unit intended to be used as a component of the M7A1 gas particulate filter unit.

6.2 Ordering data. Procurement documents should specify the title, number, date of this specification, and level of packaging and packing.

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6.3 International interest. Certain provisions of this specification are the subject of international standardization agreement ABCA-ARMY-STD-154. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Custodian:

Army - MU

Preparing activity:

Army - MU (EA)

International interest (see section 6.3)

Project No. 5950-A097

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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Edgewood Arsenal, Md. 21010

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